Mouhamadou Lamine SOW

Computer Science Engineering Student — Data Science / Machine Learning Internship

Final-year internship • 6 months • from March 2027

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Profile

Final-year engineering student at Sup Galilée, specializing in **data science and machine learning**: predictive modeling, deep learning and computer vision. Built a network-risk prediction model at **Orange (Sonatel)**; projects delivered with rigorous evaluation and reproducible code.

Professional Experience

Machine Learning / Data Science Intern May 2026 – Jul 2026
Orange (Sonatel) Dakar, Senegal

- Designed an XGBoost model predicting network risks 6 months ahead: **88 % fewer missed at-risk cases** than the existing business rule (PR-AUC 0.81 vs. 0.28).
- Built, with a teammate, an **8-year fiber network history** (2018–2026); engineered temporal and geographic features with leakage-free chronological evaluation.
- Calibrated probabilities and explained predictions (SHAP); extended to **4 horizons from 3 to 12 months** and mapped risk by area to support network planning.

Job-Shadowing Internship — Digital Technology and DevOps Jul 2024 – Sep 2024
Orange (Sonatel) Dakar, Senegal

- Wrote a literature review on **intelligent DevOps automation**; introduction to Angular, LoopBack and MongoDB.

Projects

DermTriage — Dermatological Image Classification 2026

Tools: Python, PyTorch, torchvision, scikit-learn, pytest

- Classified **10,015 images into 7 classes** (HAM10000) under a leakage-free protocol: lesion-grouped splits and a held-out test set.
- Fine-tuned a ResNet-18 network: **2.5× better** than a classical baseline (macro-F1 0.66 vs. 0.26, 5-fold cross-validation).
- Ran ablation studies to measure each design choice (**+0.15 macro-F1** from deeper fine-tuning); reproducible pipeline, 55 pytest tests.

ROADEF/EURO 2020 Challenge — RTE Maintenance Scheduling 2026

Tools: Python, Gurobi, MILP, simulated annealing, local search

- Designed a greedy repair heuristic followed by simulated annealing: **100 % feasible solutions**, verified with the official checker.
- Reached solutions **within 0.52 % on average of the best known result** of this international challenge, on 12 test instances (vs. 2.83 % for the greedy baseline).

Education

Engineering Degree in Computer Science — Master's level, expected 2027 Sep 2024 – Present
Sup Galilée — Université Sorbonne Paris Nord Villetaneuse, France

Key coursework: machine learning, deep learning, image processing, text mining, operations research.

Integrated Preparatory Program (CPI) — MPSI Sep 2022 – May 2024
ESIGELEC–ESMT Dakar, Senegal

Mathematics, physics and engineering sciences.

Skills

Programming	Python, SQL, C/C++
Machine Learning	scikit-learn, XGBoost, pandas, NumPy; imbalanced classification, cross- and time-based validation, calibration, SHAP
Deep Learning & Vision	PyTorch, torchvision, OpenCV, scikit-image; CNNs, fine-tuning, mathematical morphology
NLP	spaCy; preprocessing, named entity recognition, term extraction
Optimization	MILP, Gurobi, PuLP; metaheuristics
Tools	Git, Linux, pytest, Jupyter

Languages & Certification

French: native **English:** B2 — TOEIC 802

IBM AI Engineering Professional Certificate — IBM / Coursera, in progress (2026).